

EC466: Estimation and Inference in Econometrics

Worksheet # 2

Let $\mathbf{y}$ be an $n \times 1$ vector, $\mathbf{X} = \mathbf{1}_n$ be an $n \times 1$ vector of ones, β is a scalar, and ε is $n \times 1$ vector of errors. We write a regression equation in the vector form as follows

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon.$$

1. Write down $\mathbf{X}'\mathbf{X}$? What are the numbers of rows and columns?
2. Is $\mathbf{X}'\mathbf{X}$ invertible?
3. Write down $\mathbf{X}'\mathbf{y}$. What are the numbers of rows and columns?
4. What are the numbers of rows and columns of $\mathbf{X}'\mathbf{y}$?
5. Suppose that the above variables and equation satisfy the Gauss-Markov conditions. And we estimate the above equation by the OLS estimator. Write down b as a function of y_i 's.
6. Write down residuals by using $\mathbf{e} = \mathbf{y} - \mathbf{X}\mathbf{b}$ formula.
7. Calculate the average of the residuals. Explain why your answer makes sense or why it does not make sense.